

The idiot box

Time for the telly. This will only work if you have a normal analogue cable connection, not if you have DTH. Anyway, let's go...

First of all, install the essential build packages:

```
sudo apt-get install build-essentials linux-headers
```

Now, download the tuner drivers from <http://linuxtv.org/repo/>, and extract the tarball somewhere. Get into the directory from a terminal and issue the following code:

```
make
sudo make install
```

Notice that there is no configure step here.

Now, download the card firmware from <http://steventoth.net/linux/hvr1800/>. Download at least the zip file and the extraction shell script, and run the script. Two .fw files should be extracted. Copy these to `/lib/firmware`.

Now reboot the machine to achieve a clean state, and then run the following code:

```
sudo modprobe (cx23885, tuner)
```

If this stage goes well (you should investigate the `dmesg` output), the tuner now officially works under Linux. The next step is installing MythTV and integrating it with XBMC, so you can use your XBMC interface to watch and record live TV.

Type the following command to download and install MythTV:

```
sudo apt-get install mythtv
```

First, MythTV needs to be configured. No instructions, sorry! There're too many variables to be considered here. The device node for your TV tuner card is `/dev/video0`. To watch TV, you need to manually scan the line for all frequencies that have channels (the frequencies depend on your local cable company). Then you need to key in channel numbers and names manually. Getting hold of an EPG (Electronic Program Guide—the stuff that shows schedules) is also difficult, but you can find some on MSN, which I'm not sure is MythTV-compatible. You need an EPG though—or else Mythbox won't work.

Use this Web resource to help configure MythTV: http://parker1.co.uk/mythtv_ubuntu.php. Although it's not for India, most of the stuff described in there does apply. And it's a long document, mind you.

After you have done all the configuration in MythTV (make sure it works exactly as you intend—you need to be able to watch channels and use the EPG perfectly), it's time to hook up MythTV to XBMC, and turn the media centre into a TiVo as well.

Execute the following code:

```
sudo apt-get install ffmpeg python-mysqldb
```

Then, head to <http://code.google.com/p/mythbox/> and download the latest tarball. Extract it into `$HOME/.xbmc/` scripts, like:

```
cd ~/.xbmc/scripts
tar -xvzf /path/to/mythbox-svn-1260.tar.gz
```

Start up XBMC again (use `Alt+F2`, then run `xbmc -fs`; you need to be in full screen). Now go to *Scripts*→*MythBox*. You'll be sent into a *Settings* screen. Fill in your settings—mostly database settings—and if you just can't get MythBox to authenticate, use 'root' as the user name and the root password (the MySQL root password that you supplied when you were setting up MythTV). Now go to *Test Settings*. If *Settings OK* comes up, exit the settings screen (Press `Esc`), and enjoy Live TV. You can even schedule recordings. To get into MythBox, go to *Scripts*→*MythBox*. Now instead of going to the settings screen, you will be taken to the TV console. Enjoy!

Final touches

Finally, it's time to get it all started silently. First of all, go to *System*→*Administration*→*Login Window*. Type in your password, and then go to the *Security* tab. Here, enable auto-login, and select the appropriate user to log in as. This should ensure that you do not have to type in your credentials to start using the media centre.

The next thing to do is make XBMC start up automatically. Go to *System*→*Preferences*→*Startup Applications*. Hit *Add*. Type in anything in the *Name* field. Type in `xbmc -fs` in the command field. Click *Add*, and then *Close*.

Now reboot your media centre. Voila, it boots straight into XBMC!

Anything else?

Sure, why waste such a good piece of hardware? Besides running XBMC, which is using up all your GPU power, you can run an Icecast server in the background to stream your music over Wi-Fi, or use VLC to stream videos as well. This makes good use of the leftover processing power. Yes, with that 1TB hard disk, you can sure store all your H.264 movies, music and pictures, and still have some space left, so why not use it as a home server? Innovate, while I try out my TV-less XBMC set-up. (Yes, I don't have a tuner because I use Tata Sky :-)) **END** 

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The author is a 15-year-old student studying in Class 9. He is a logician (as opposed to a magician), a great supporter of Free Software and loves hacking Linux. Other than that, he is an experienced programmer in BASIC and can also program in C++, Python and Assembly (NASM Syntax).

MythBox screenshots are courtesy: <http://code.google.com/p/mythbox/wiki/Screenshots>